Hamza Khan

AI & Full Stack Engineer

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PROFESSIONAL SUMMARY

Engineer with experience in full-stack web development, machine learning research, and DevOps. Builds responsive interfaces, backend APIs, WordPress websites, and AI services using React, Laravel, Python, Docker, and AWS. Pursuing a Ph.D. in Computer Science with research focused on violence detection in industrial surveillance using deep learning.

TECHNICAL SKILLS

Frontend & CMS: React.js, JavaScript, HTML5, CSS3, responsive web design, reusable UI components, WordPress custom themes and plugins

Backend & languages: Python, PHP, Laravel, Flask, FastAPI, REST APIs, microservices, Java, C/C++, SQL, Bash

AI & data: PyTorch, TensorFlow, Scikit-learn, CNN, LSTM, model evaluation, LLM integration, prompt engineering, Hugging Face, OpenAI APIs, Pandas, NumPy, PostgreSQL, MySQL

Cloud & delivery: AWS (EC2, S3, IAM, CloudWatch, ALB), Docker, Kubernetes, Terraform, CloudFormation, Jenkins, GitHub Actions, GitLab CI, CI/CD, automated testing, monitoring, Git, Linux, Agile/Scrum

PROFESSIONAL EXPERIENCE

Graduate Research Assistant

Aug 2022 - Present

North Carolina A&T State University, Greensboro, NC

- Develop Python-based machine learning pipelines for violence detection in surveillance videos using deep learning models.
- Build automated evaluation workflows for model benchmarking, inference monitoring, and output validation across datasets.
- Integrate LLMs with prompt templates, automated response evaluation, and REST services for scalable inference.
- Build Flask inference APIs and containerized services on AWS EC2; automate deployment using GitHub Actions.

Lead Developer / Technical Project Manager

Feb 2025 - Feb 2026

EzyTeams

- Led development and delivery of SaaS collaboration tools with cross-functional engineering teams.
- Coordinated feature implementation, backlog prioritization, release planning, and resolution of technical blockers.
- Supported product demos, stakeholder communication, and alignment of engineering delivery with product roadmaps.

DevOps Engineer

Sep 2022 - Nov 2024

Byte Forge Information Technology Services, Dubai, UAE

- Built application build, test, and deployment pipelines with Jenkins, GitHub Actions, and GitLab CI.
- Provisioned AWS infrastructure using Terraform and CloudFormation; deployed Python applications and ML microservices with Docker and Kubernetes.
- Automated logging and monitoring workflows, debugged production issues, and optimized system performance.

Full Stack Developer May 2021 - May 2022

Pakistan Professional Development Academy, Haripur, Pakistan

- Developed and redesigned the Inspire London College website, improving user experience and mobile responsiveness.
- Built web applications using Laravel, JavaScript, and React with REST API integrations.
- Created custom WordPress themes and plugins to simplify content management for non-technical users.

Frontend Developer Oct 2020 - Apr 2021

Virtelligence, Pakistan

- Developed React.js frontend interfaces integrated with Laravel-based backend services.
- Collaborated with backend developers to connect React components to Laravel APIs for dynamic data rendering.
- Built responsive, reusable UI components to maintain consistent design across application views.

SELECTED PROJECTS

Deep Learning Model Deployment Pipeline

Designed an end-to-end pipeline for training, packaging, and serving models. Built Flask inference APIs, deployed Docker containers on AWS EC2, and automated model updates using GitHub Actions.

Violence Detection in Surveillance Videos

Built a TensorFlow/Keras 3D CNN video-classification pipeline using four Conv3D blocks and 16-frame RGB plus optical-flow clips. Automated data splitting, preprocessing, training, and checkpoint selection; evaluated video-level predictions using aggregation methods, threshold analysis, accuracy, precision, recall, F1, and AUC.

Human-Centric Security in AI-Enhanced CPS

Analyzed human behavioral vulnerabilities and interactions between software control logic and physical components. Proposed mitigation strategies for AI-integrated cyber-physical systems.

Inspire London College Website

Developed during my role at Pakistan Professional Development Academy. Improved mobile responsiveness and content management through WordPress themes and plugins.

EDUCATION

Ph.D. in Computer Science (in progress) | 2024 - Present

North Carolina A&T State University, Greensboro, NC | GPA: 3.90

M.S. in Computer Science | Aug 2022 - Oct 2023

North Carolina A&T State University, Greensboro, NC | GPA: 3.88/4.00

B.S. in Software Engineering | Aug 2016 - Oct 2020

University of Haripur, Haripur, Pakistan | GPA: 3.90/4.00

CERTIFICATIONS

IBM DevOps (Coursera) | AWS Certified Cloud Practitioner (in progress)